DIENCEPHALON - I

THALAMUS

LEARNING OBJECTIVES

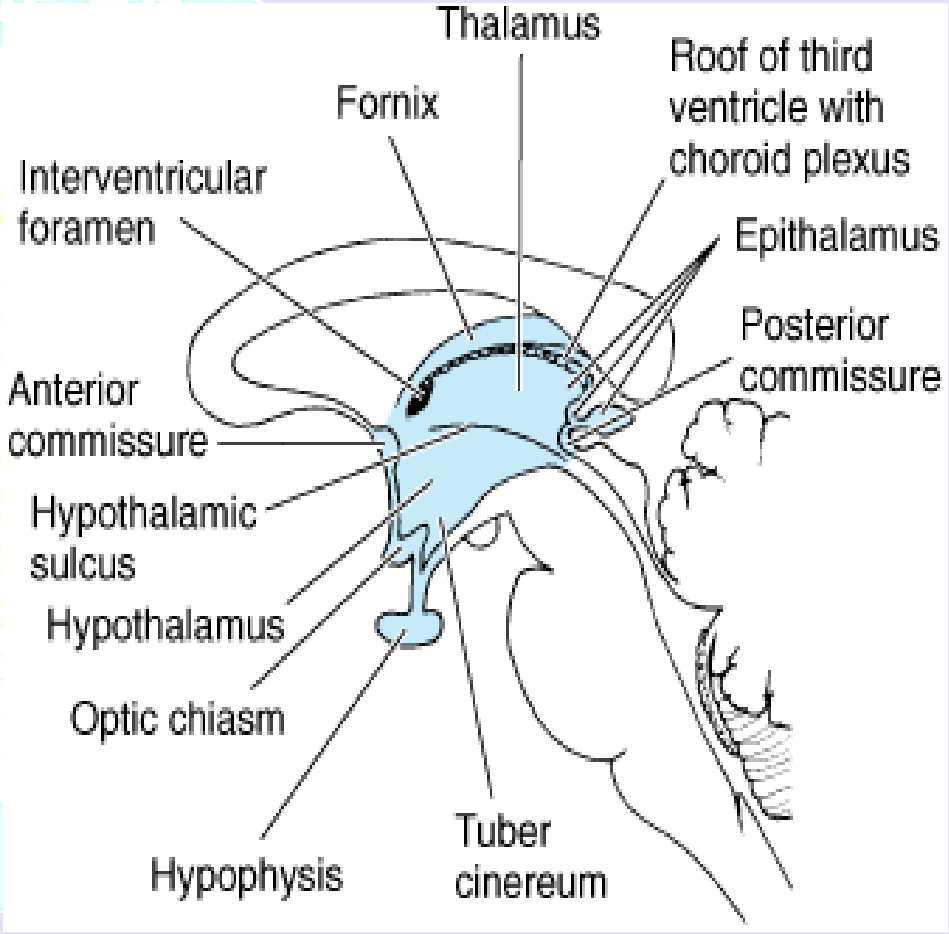
At the end of this lecture, students should be able to:-

- Know the structure and divisions of Diencephalon
- Understand the boundaries of diencephalon
- Narrate the fuctions, nuclei and connections of THALAMUS

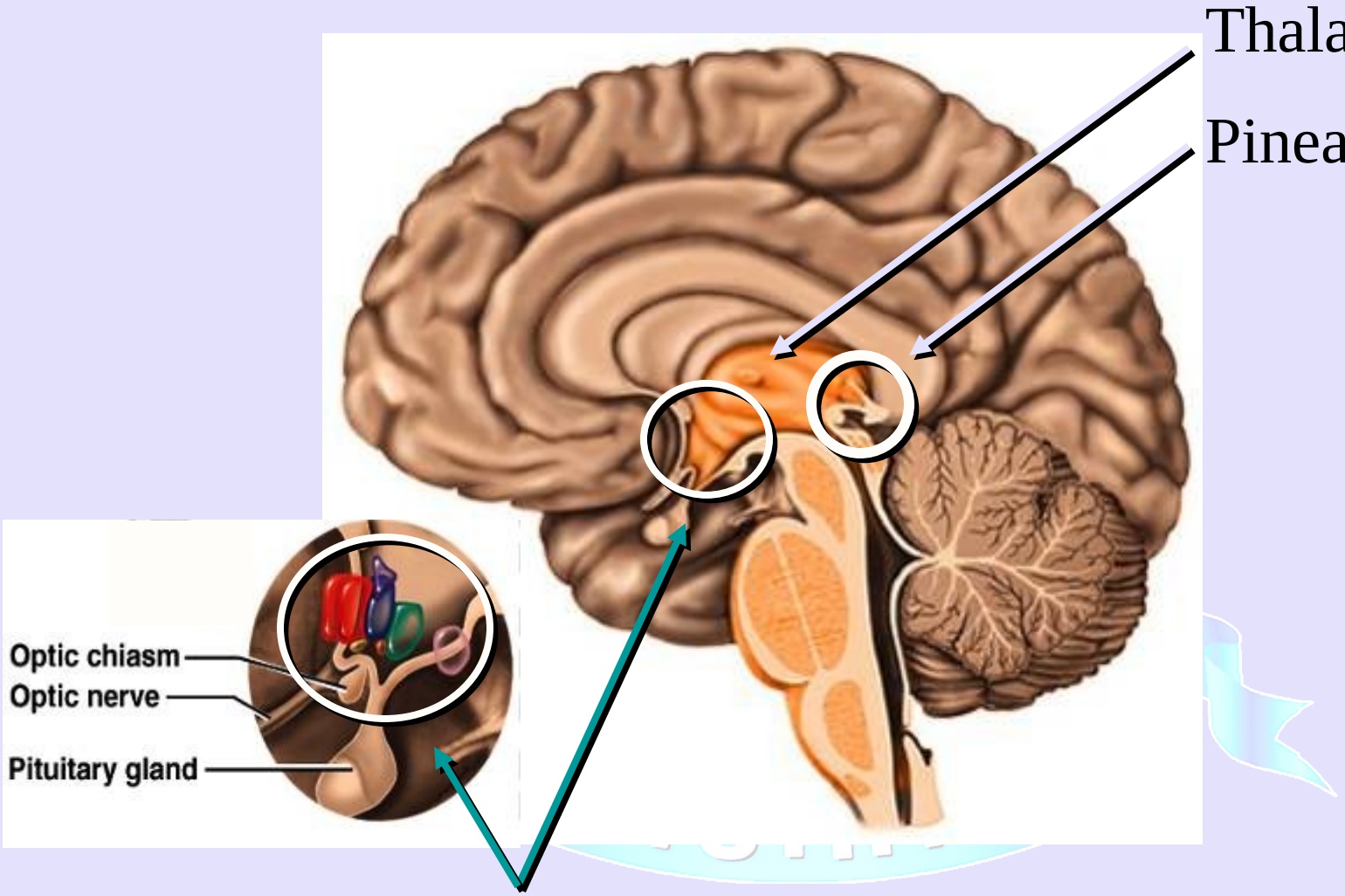
DIENCEPHALON DIVISIONS

- THALAMUS
- HYPOTHALAMUS
- EPITHALAMUS
- SUBTHALAMUS
- The third ventricle lies between the halves of the diencephalon.

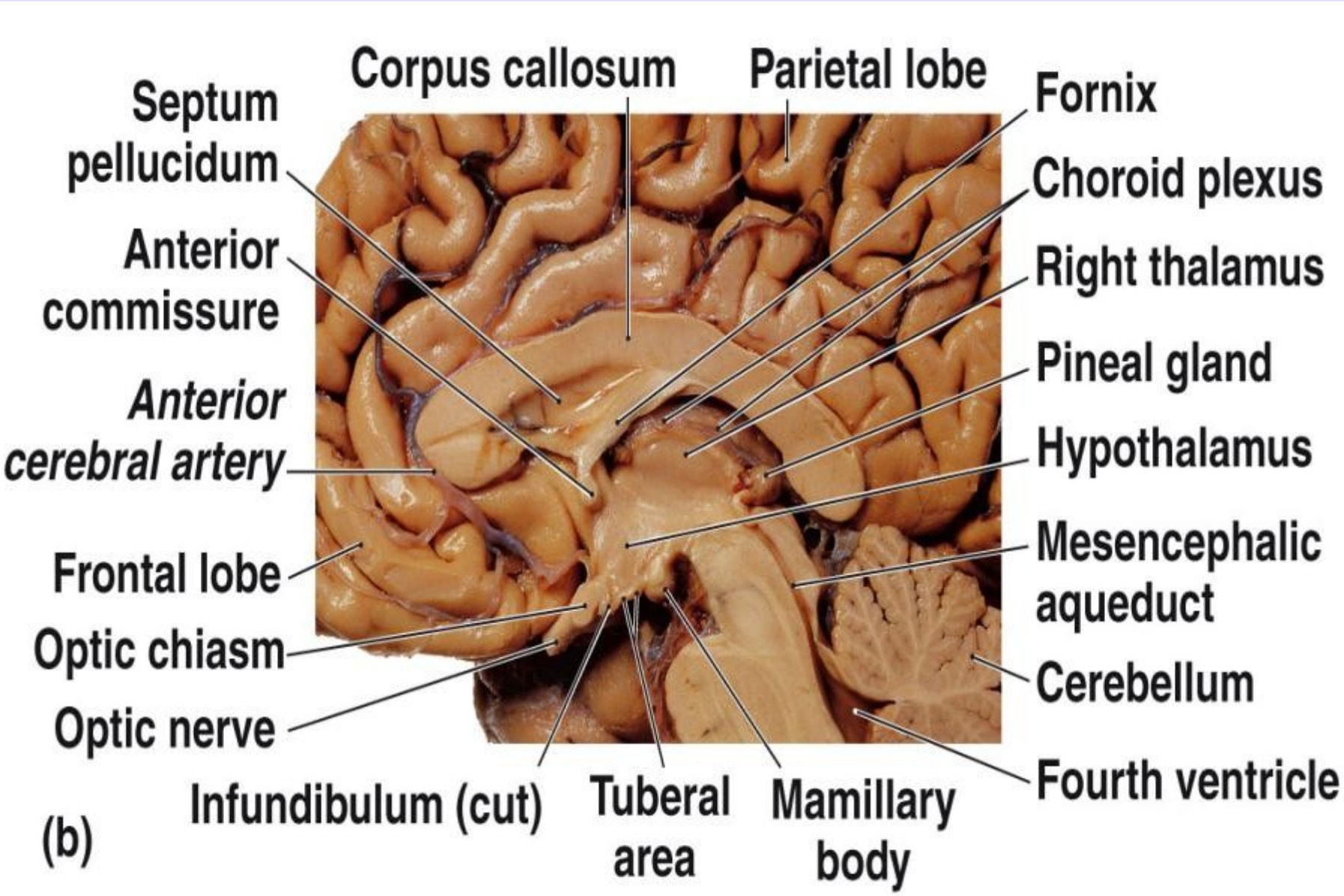
- **FUNCTIONS**
- important for regulation of basic functions and linkage of basic functions to more complex functions such as movement



DIENCEPHALON

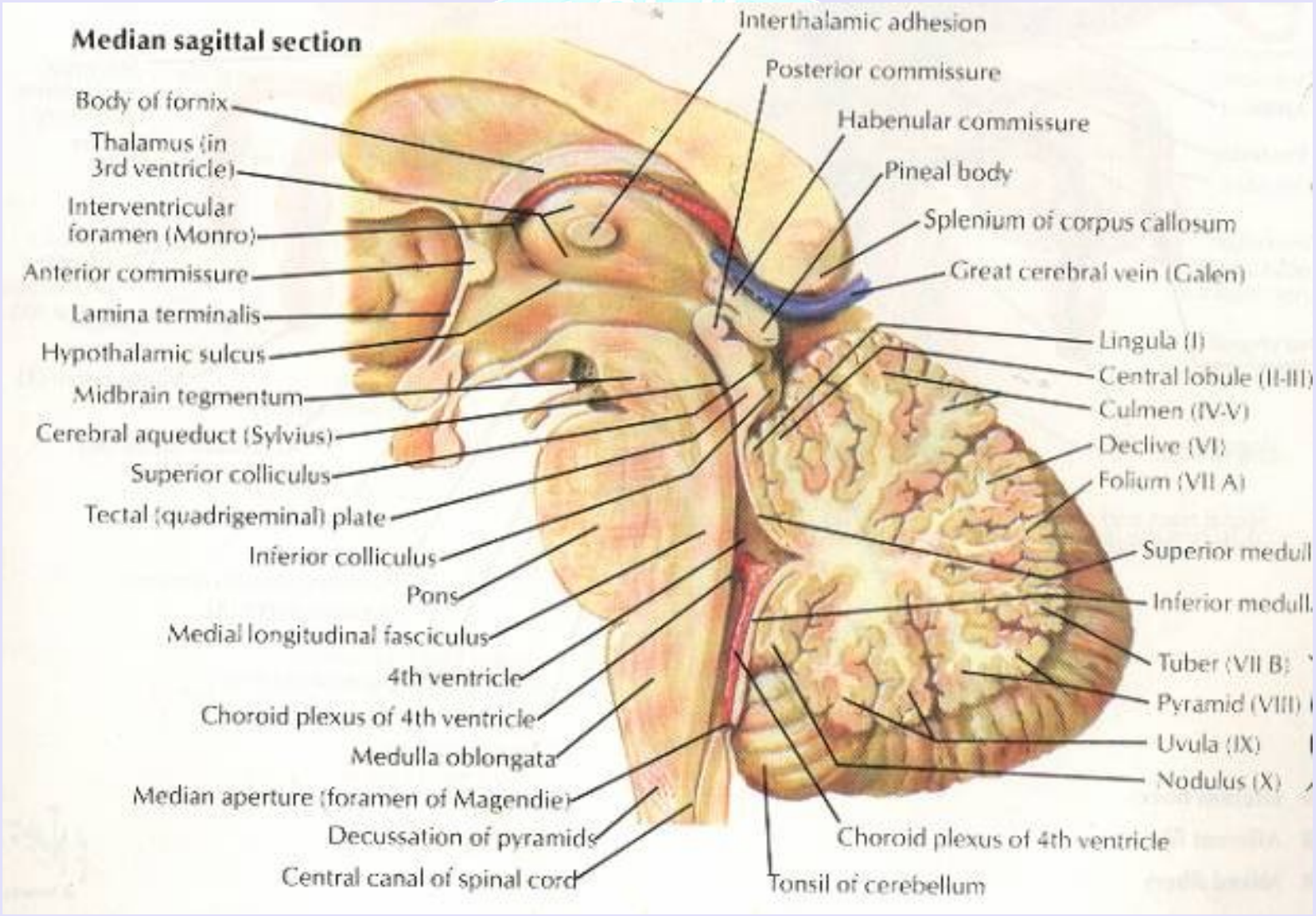


Thalamus, Hypothalamus and Epithalamus (houses pineal gland)

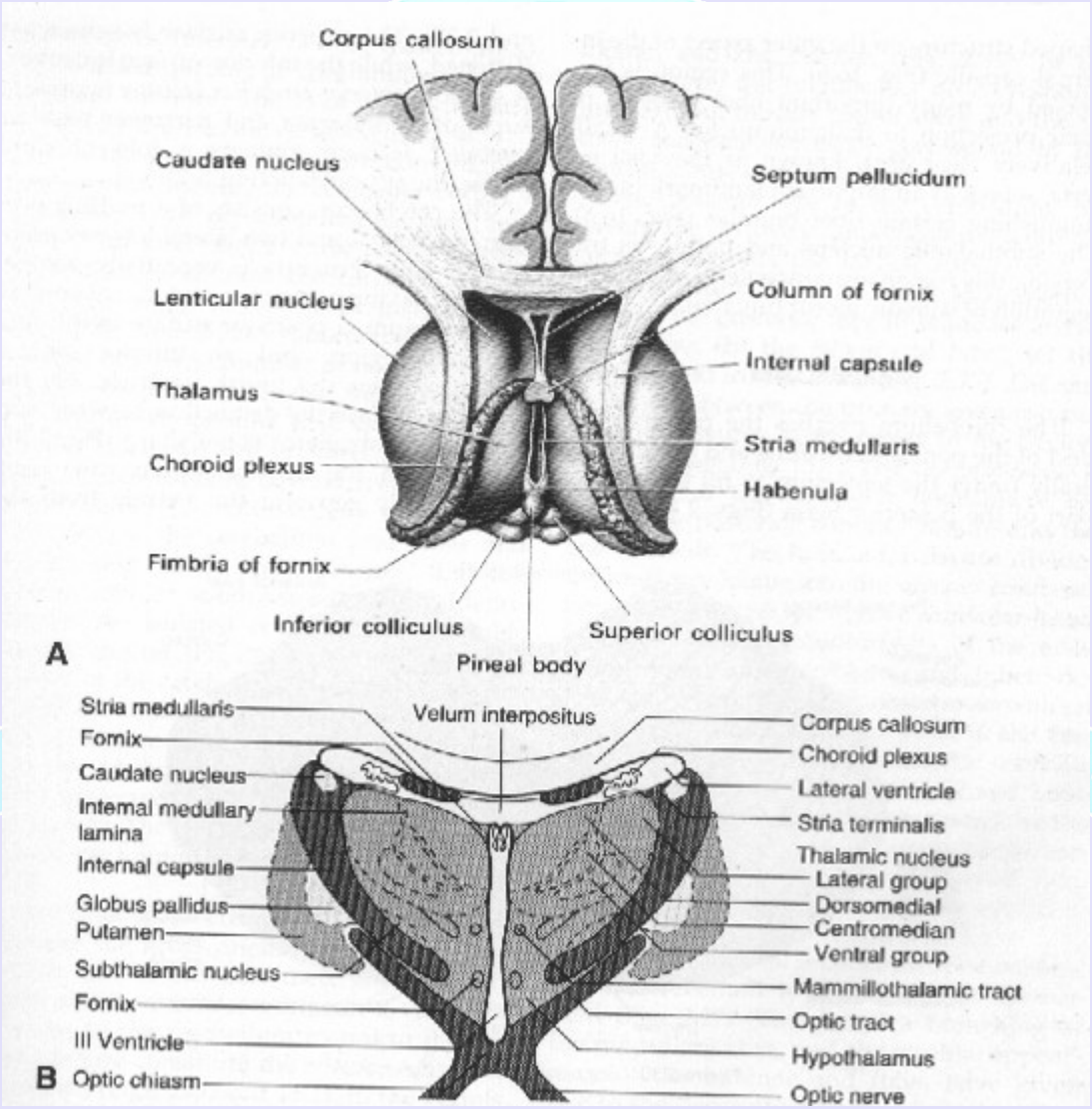


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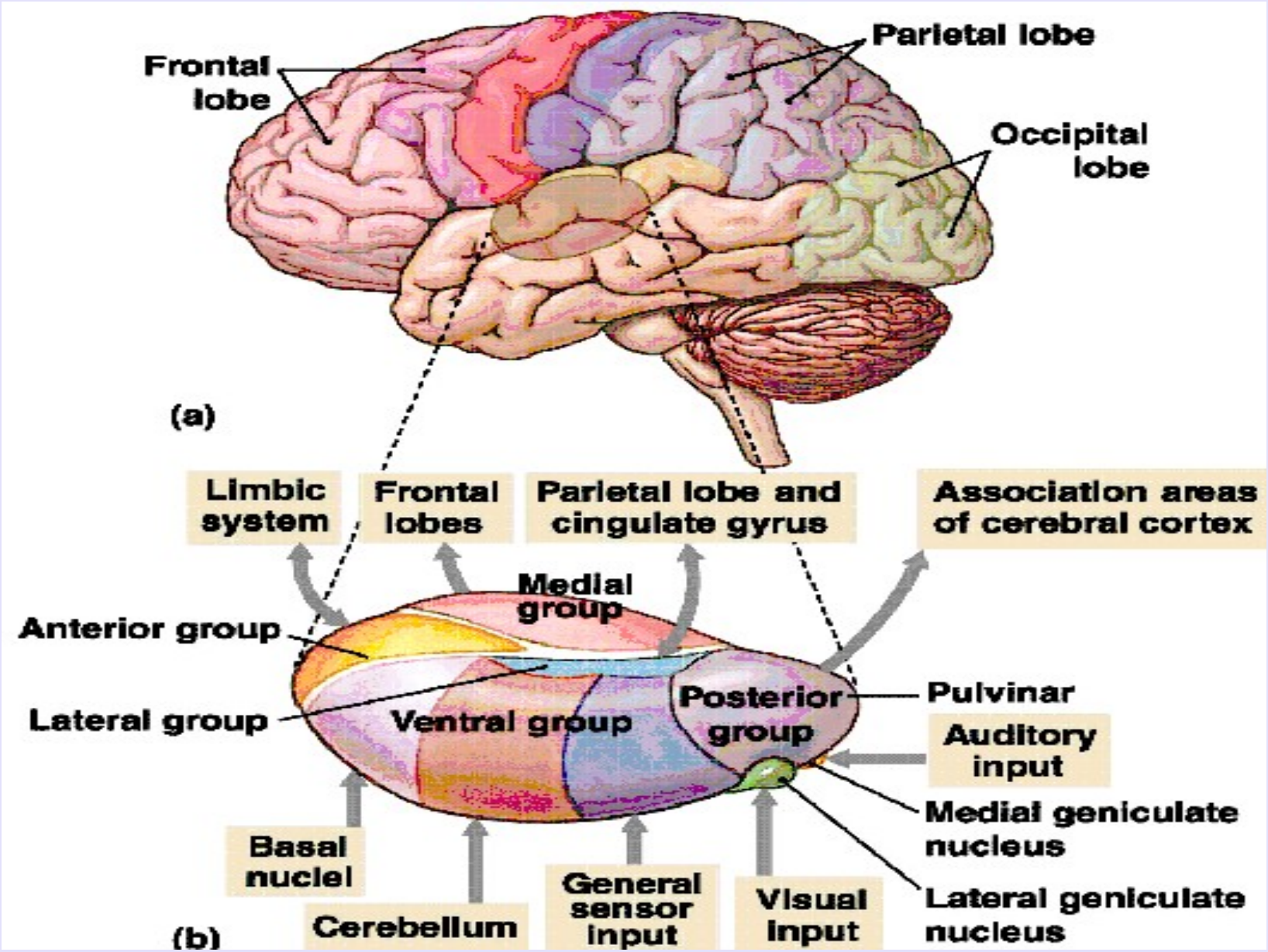
DIENCEPHALON BOUNDARIES



DIENCEPHALON BOUNDARIES



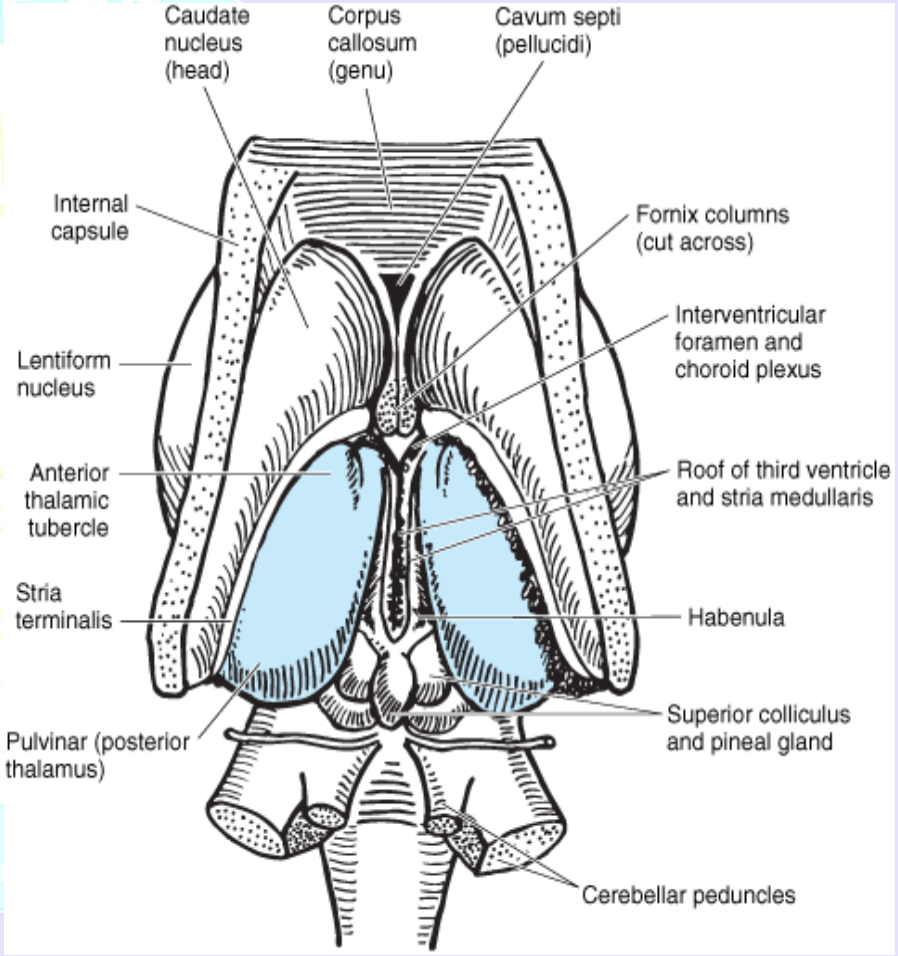
THALAMUS


Frontal lobe
Parietal lobe
Occipital lobe
(a)
Limbic system
Frontal lobes
Parietal lobe and cingulate gyrus
Association areas of cerebral cortex
Medial group
Anterior group
Lateral group
Ventral group
Posterior group
Pulvinar
Auditory input
Medial geniculate nucleus
Lateral geniculate nucleus
Basal nuclei
Cerebellum
General sensor input
Visual input
(b)

THALAMUS

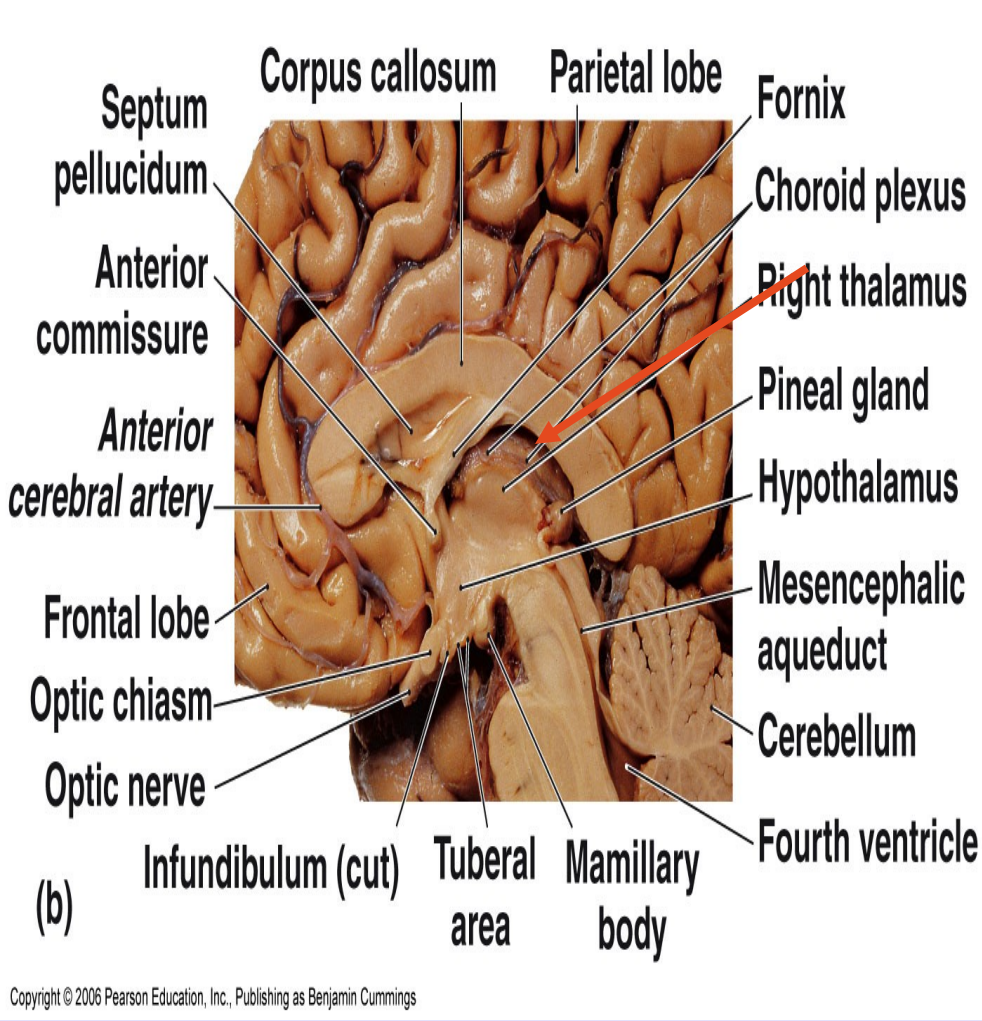
Intermediate mass

- Several nuclei:
- Major relay station for most sensory impulses.
- The left thalamus and right thalamus are separated by the third ventricle.
- A projection of gray matter called the intermediate mass extends into the ventricle from each side.



THALAMUS - FUNCTIONS

- Distributing most of afferent inputs to cerebral cortex
- Control of electrocortical activity of cerebral cortex – plays important roles in arousal, consciousness and sleep mechanisms
- Integration of motor functions by providing the relays – impulses from the basal ganglia and cerebellum can reach the motor cortex

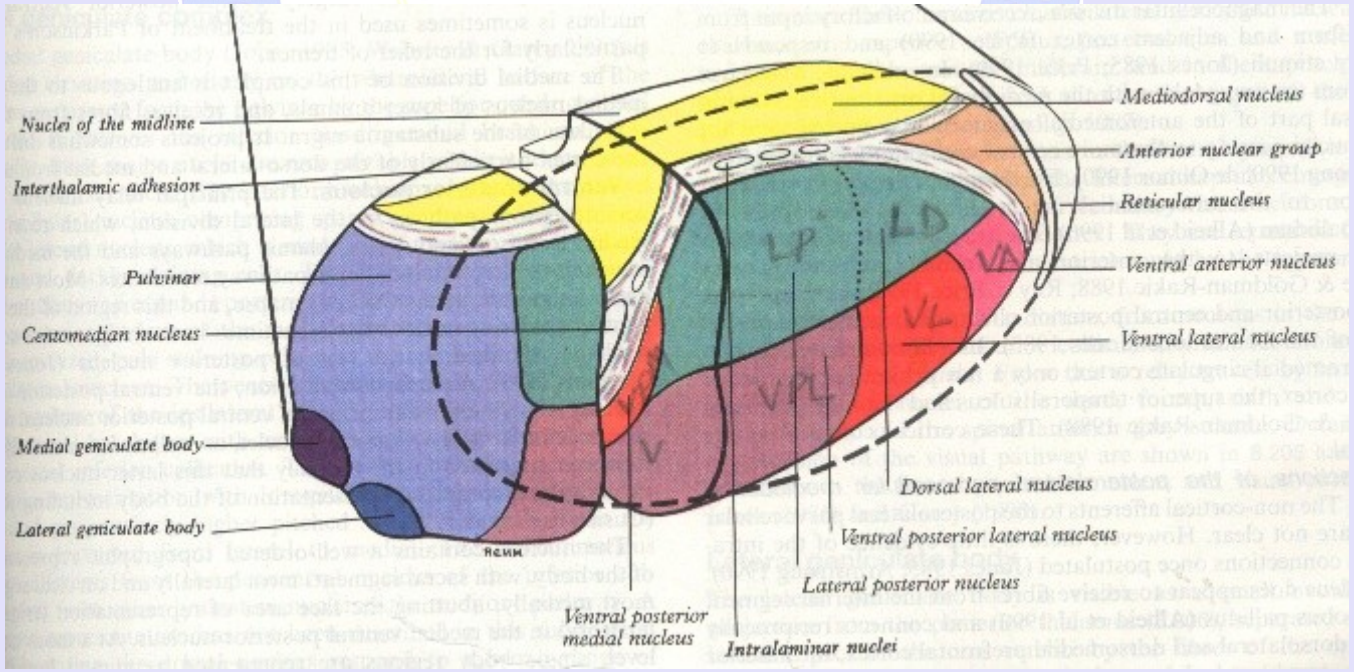


THALAMUS DIVISIONS

- DORSAL THALAMUS OR THALAMUS PROPER
- VENTRAL THALAMUS
  - Reticular nucleus
  - Ventral lateral geniculate nucleus
  - Zona incerta
  - Nucleus of field of Forel
- EPITHALAMUS

DORSAL THALAMUS

- **DIVISIONS**
  - **Three nuclear masses – internal medullary lamina**
  - **ANTERIOR**
  - **LATERAL**
  - **MEDIAL**
  - **OTHERS**
    - **POSTERIOR**
    - **INTRALAMINAR**
    - **MEDIAN- MIDLINE NUCLEI**
    - **GENICULATE NUCLEAR GROUP - METATHALAMUS**



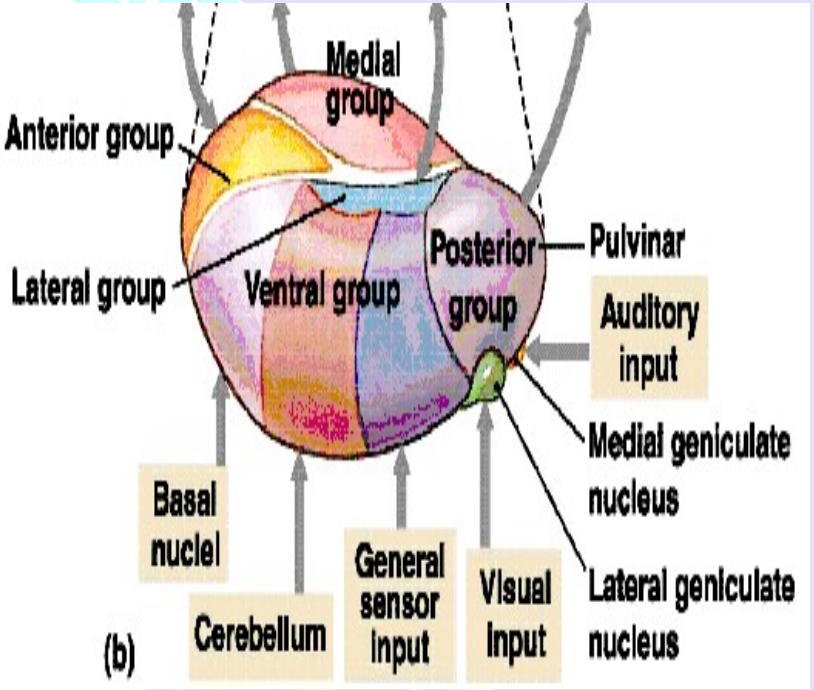
THALAMUS
CONNECTIONS

- **The dorsal thalamic nuclei project to and receive fibers from cerebral cortex**
- **Topographically organized**
- **Depending upon connections thalamic nuclei can be classified as:**
  - ***Specific relay nuclei- R***
  - ***Association nuclei- A***
  - ***Nuclei with subcortical projections- SC***
  - ***Nuclei with diffuse cortical connections- DC***

THALAMUS CONNECTIONS

- ANTERIOR NUCLEAR GROUP
- NUCLEI
  - Anteroventral- R
  - Anterodorsal- R
  - Anteromedial- R

- CONNECTIONS
  - Hippocampal formation- direct- fornix
  - Indirectly- mammillothalamic tracts
  - cingulate gyrus
  - Brainstem nuclei
  - Anterior limbic area- corpus callosum



- FUNCTIONS
- Regulation of alertness
- Acquisition of information- in attention & in encoding memories

THALAMUS
CONNECTIONS

MEDIAL NUCLEAR GROUP

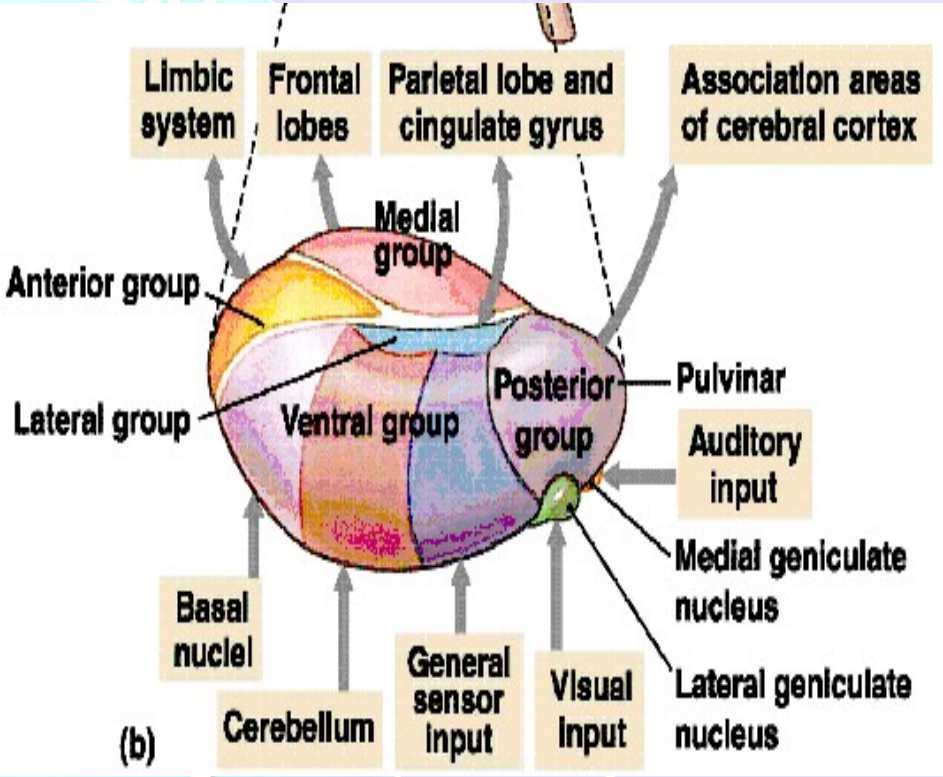
- MEDIODORSAL NUCLEUS
  - Parvicellular part-A
  - Magnocellular part- SC
  - Paralaminar part

CONNECTIONS

- Prefrotal cortex
- Amygdala
- Globus pallidus
- Olfactory input from pyriform cortex
- Cingulate, parietal,superior temporal & insular cortex

FUNCTIONS

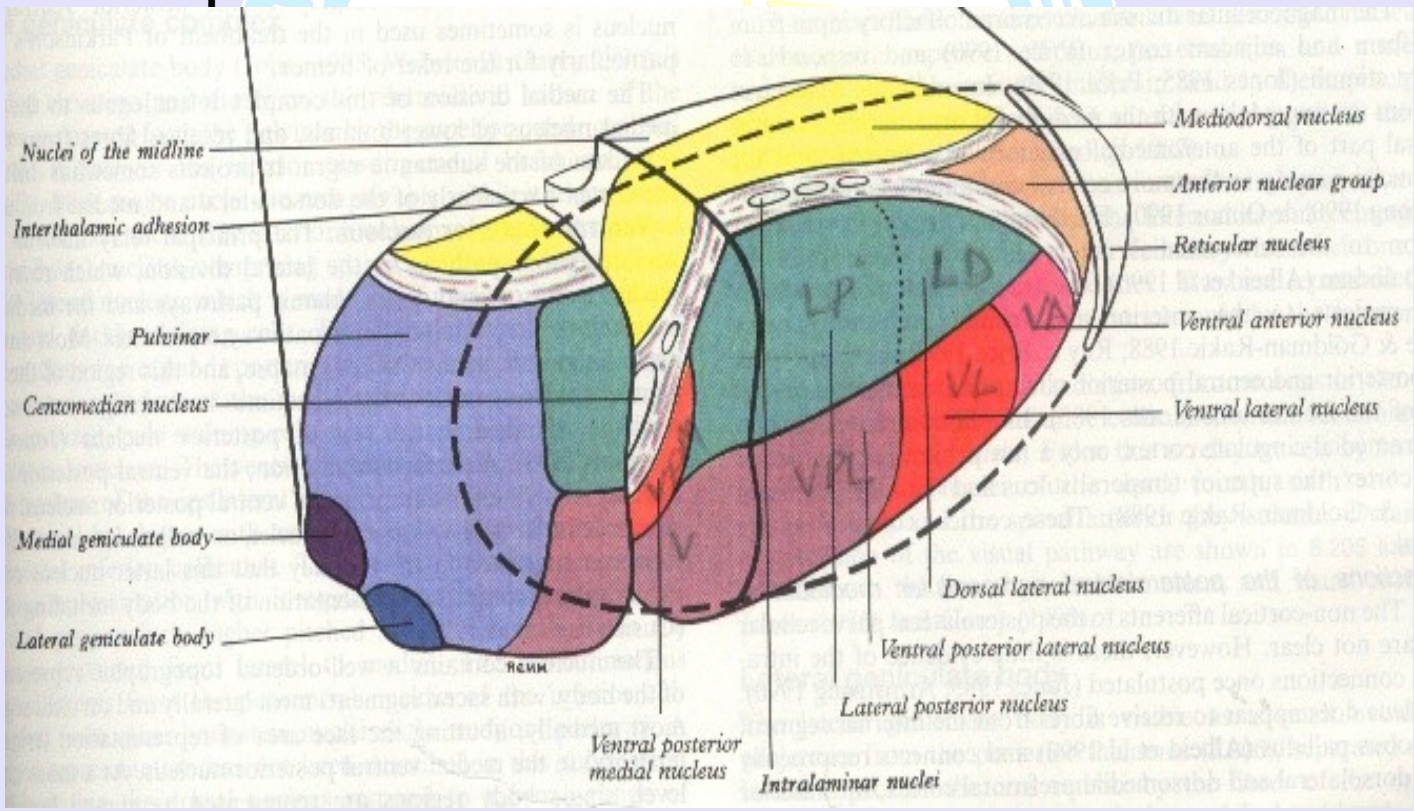
- Higher mental functions, damage causes decrease in anxiety, tension, aggression or obsessive thinking & transient amnesia



THALAMUS
CONNECTIONS

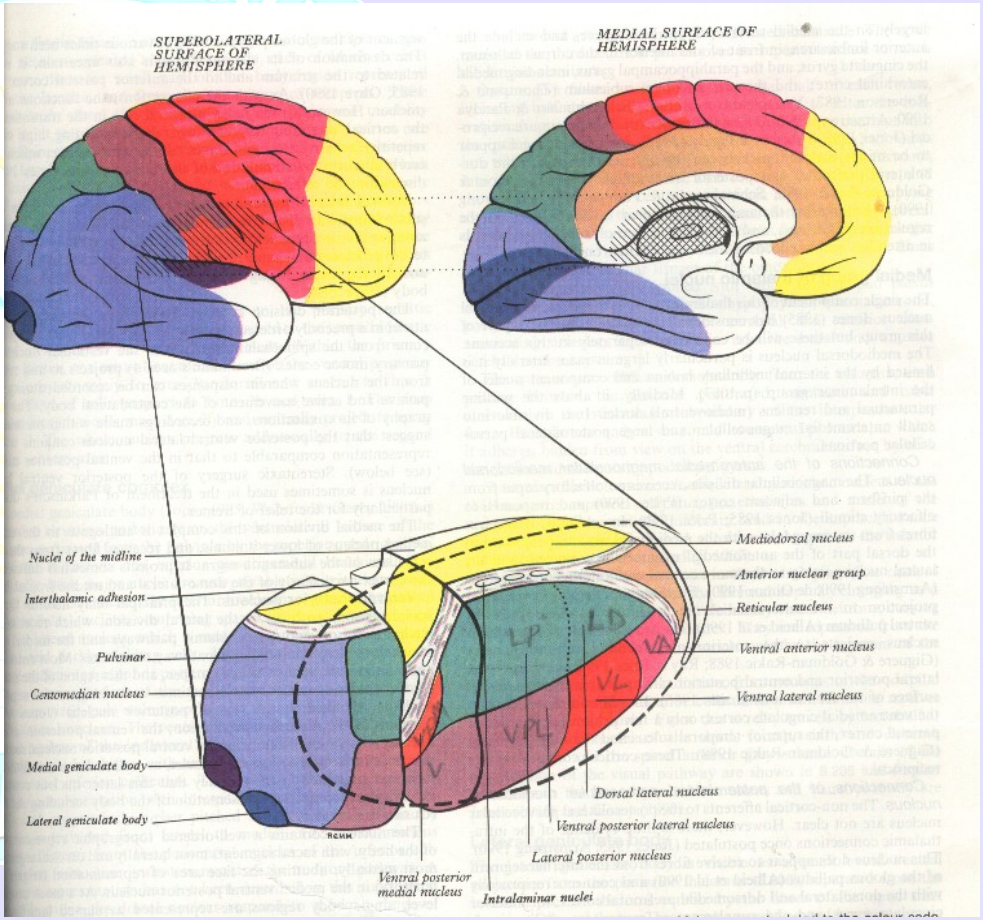
VENTRAL NUCLEAR GROUP

- NUCLEI
- Ventral anterior nucleus
- Ventral lateral nucleus
- Ventral posterior nucleus



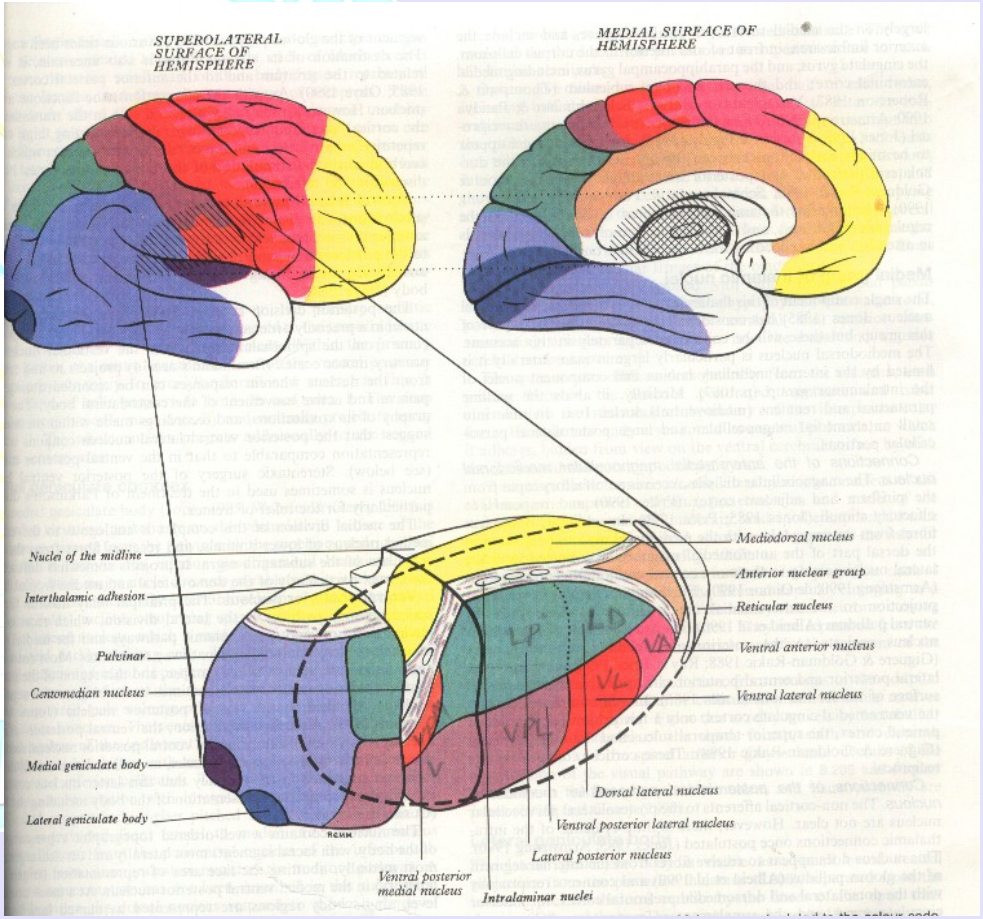
VENTRAL NUCLEAR GROUP

- **VENTRAL ANTERIOR NUCLEUS**
  - **Parvicellular part-R**
  - **Magnocellular part- R**
- **AFFERENTS**
  - **Deep cerebellar nuclei**
  - **Globus pallidus**
  - **Substantia nigra**
- **EFFERENTS**
  - **Frontal striatum**
  - **Premotor cortex**
- **FUNCTIONS**
  - **influences activity of motor cotex**



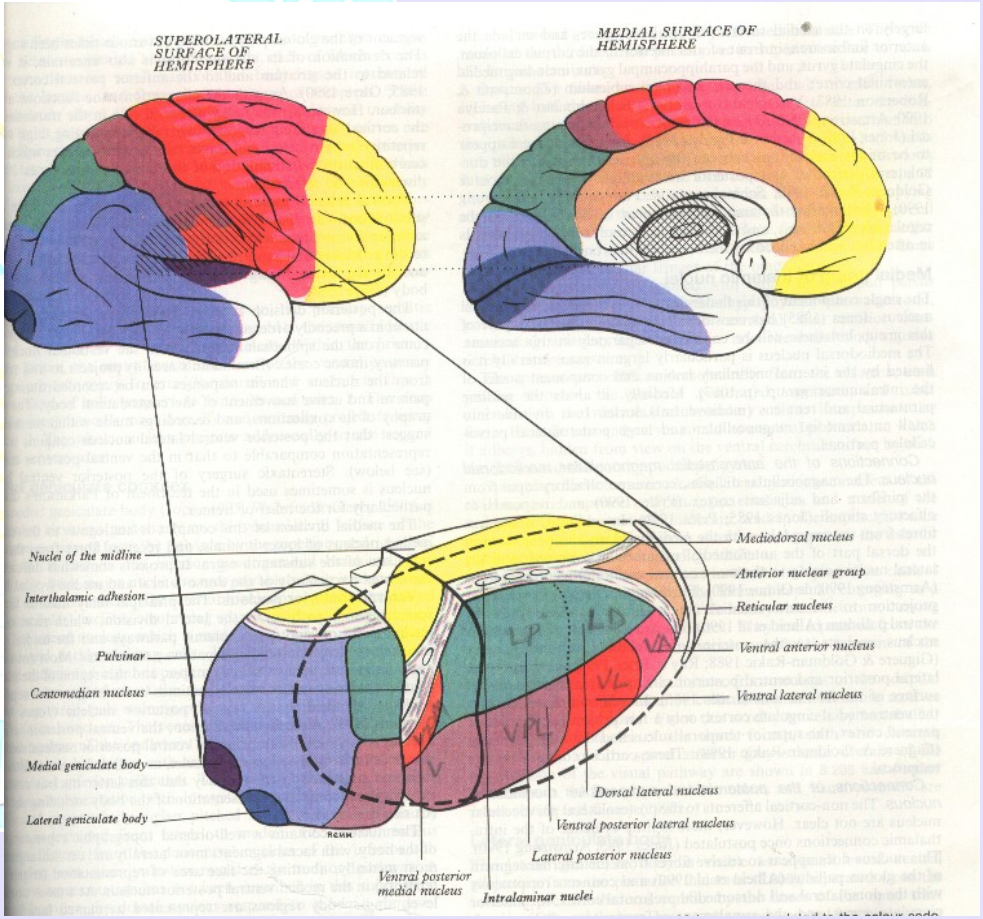
VENTRAL NUCLEAR GROUP

- **VENTRAL LATERAL NUCLEUS**
  - **Oral part- R**
  - **Caudal part- R**
  - **Medial part- R**
- **AFFERENTS**
  - **Globus pallidus**
  - **Deep cerebellar nuclei**
  - **Spinothalamic tracts**
  - **Vestibular nuclei**
  - **Substantia nigra**
- **EFFERENTS**
  - **Premotor & supplementary motor cortex**
- **FUNCTIONS**
  - **Initiation of voluntary movements**



VENTRAL NUCLEAR GROUP

- VENTRAL POSTERIOR NUCLEUS
- Largest part
- Nuclei
  - Ventral posterolateral
    - Oral part- R
    - Caudal part- R
  - Ventral posteromedial
    - Parvicellular part- R
    - Ventral posterior inferior part- R
- FUNCTION
- Largest primary somatic sensory relay nucleus
- Referred to as ventrobasal complex

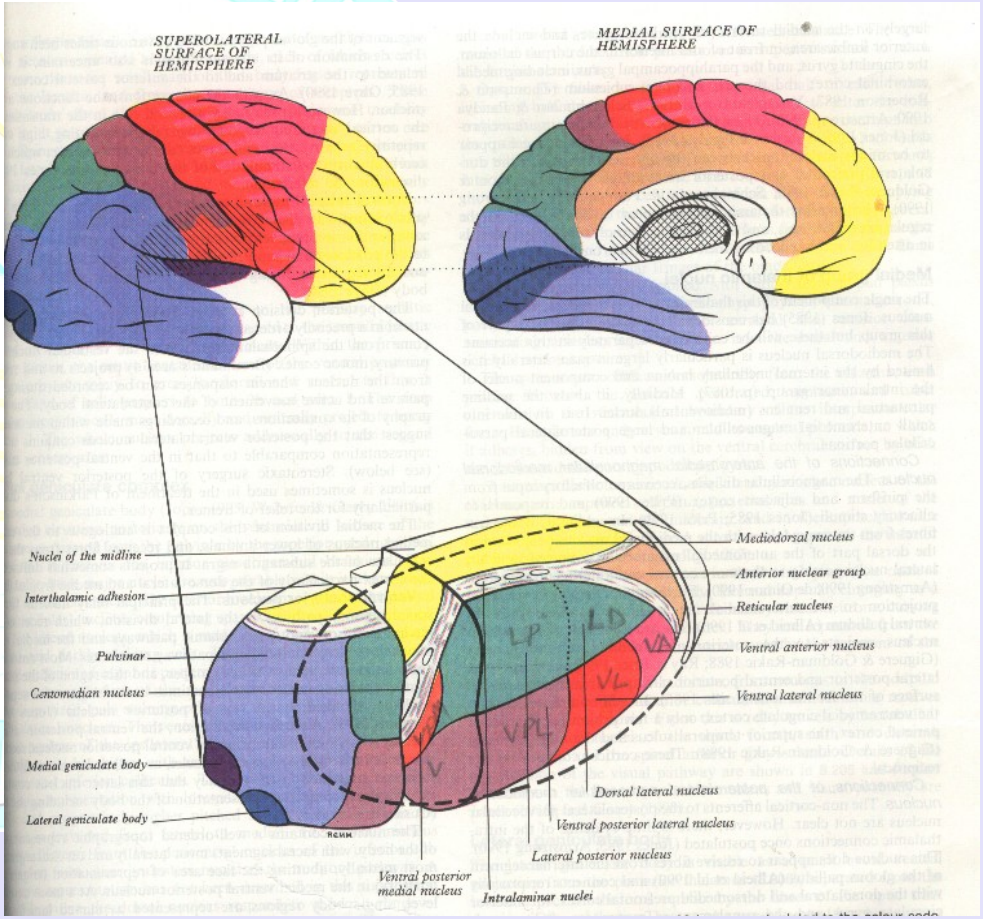


VENTRAL NUCLEAR GROUP

- **VENTRAL POSTERIOR NUCLEUS**

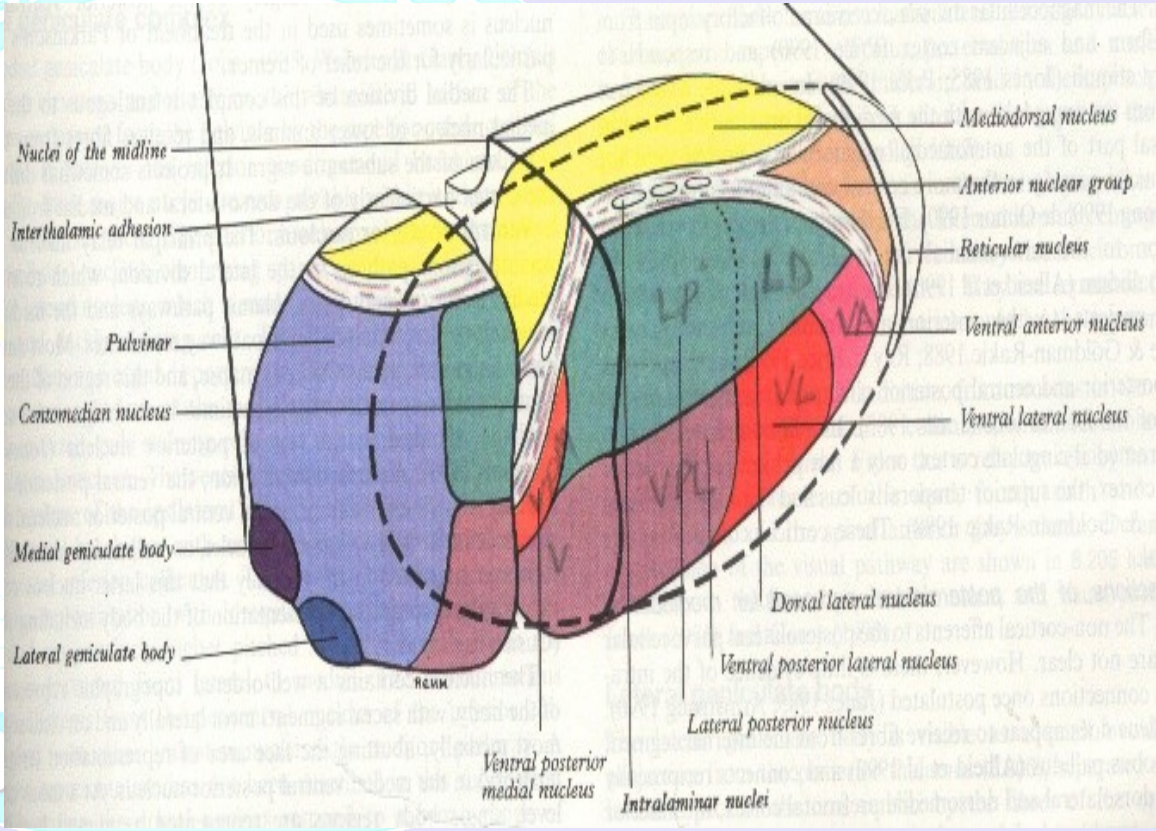
- **AFFERENTS**
  - medial leminiscus
  - spinal leminiscus
  - trigeminothalamic pathways
  - gustatory fibres

- **EFFERENTS**
  - Primary sensory cortex
  - Second somatic sensory cortex



LATERAL NUCLEAR GROUP

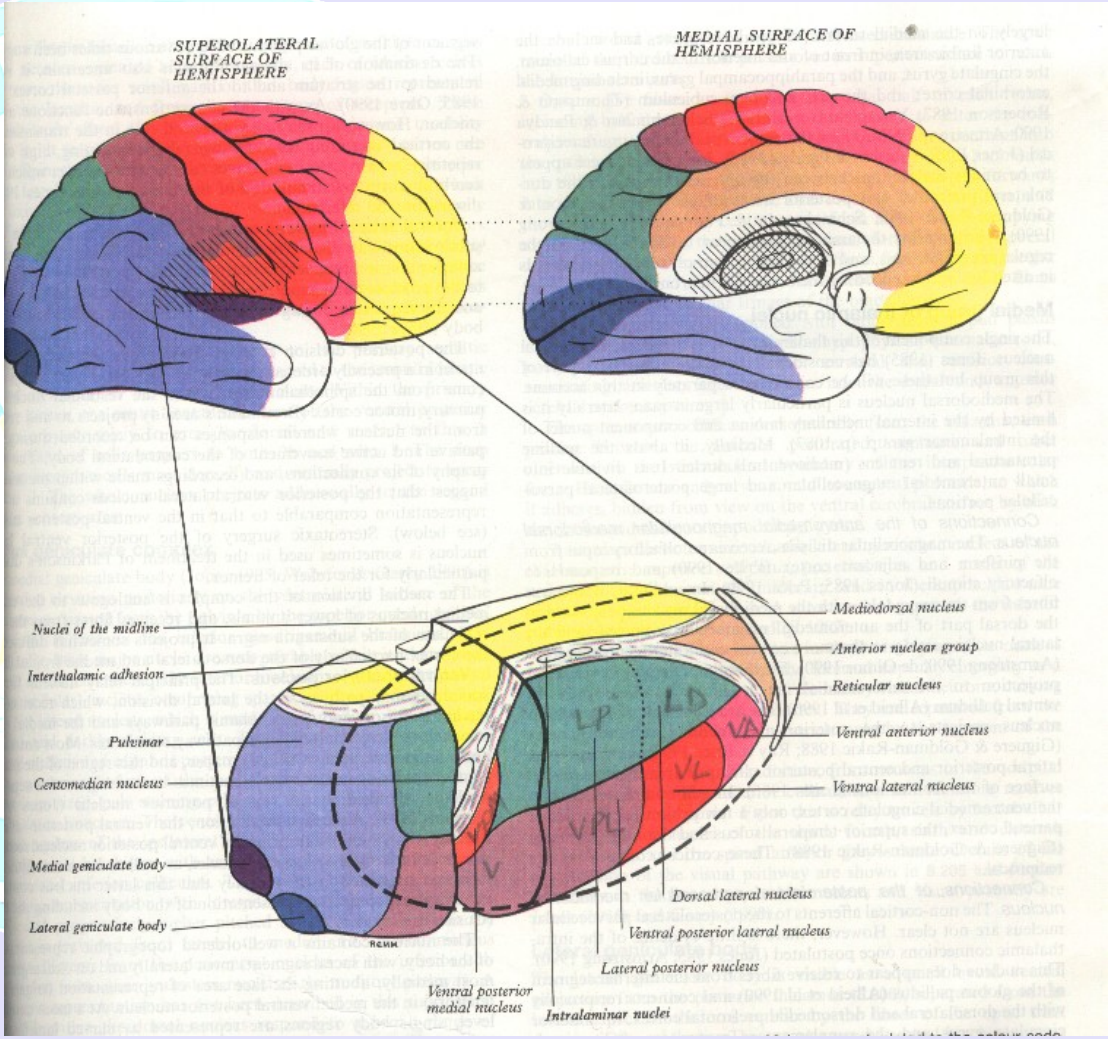
- **LATERAL DORSAL NUCLEUS- R**
- **LATERAL POSTERIOR NUCLEUS- R**
- **PULVINAR- R**
  - **Medial part**
  - **Lateral part**
  - **Inferior part**



LATERAL NUCLEAR GROUP

- **LATERAL DORSAL NUCLEUS**
- **LATERAL POSTERIOR NUCLEUS**

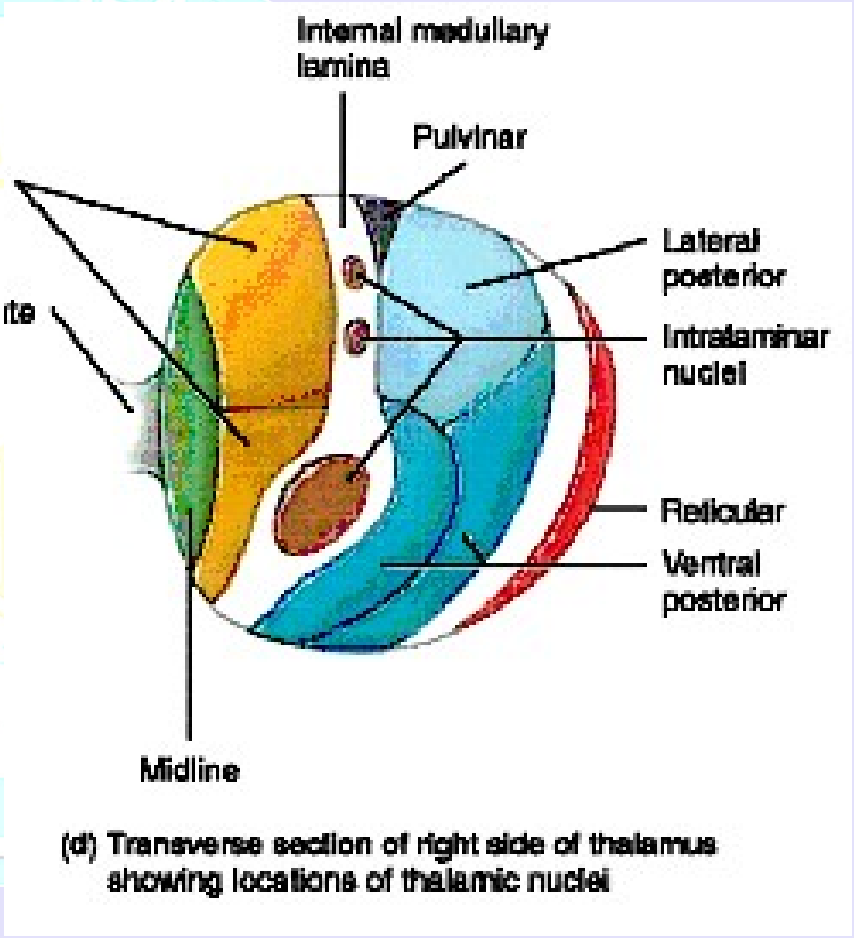
- AFFERENTS
  - **Pretectum**
  - **Superior colliculus**
- EFFERENTS
  - **Cingulate cortex**
  - **Hippocampal cortex**
  - **Parahippocampal cortex**
  - **Parietal cortex**



LATERAL NUCLEAR GROUP

- **PULVINAR**

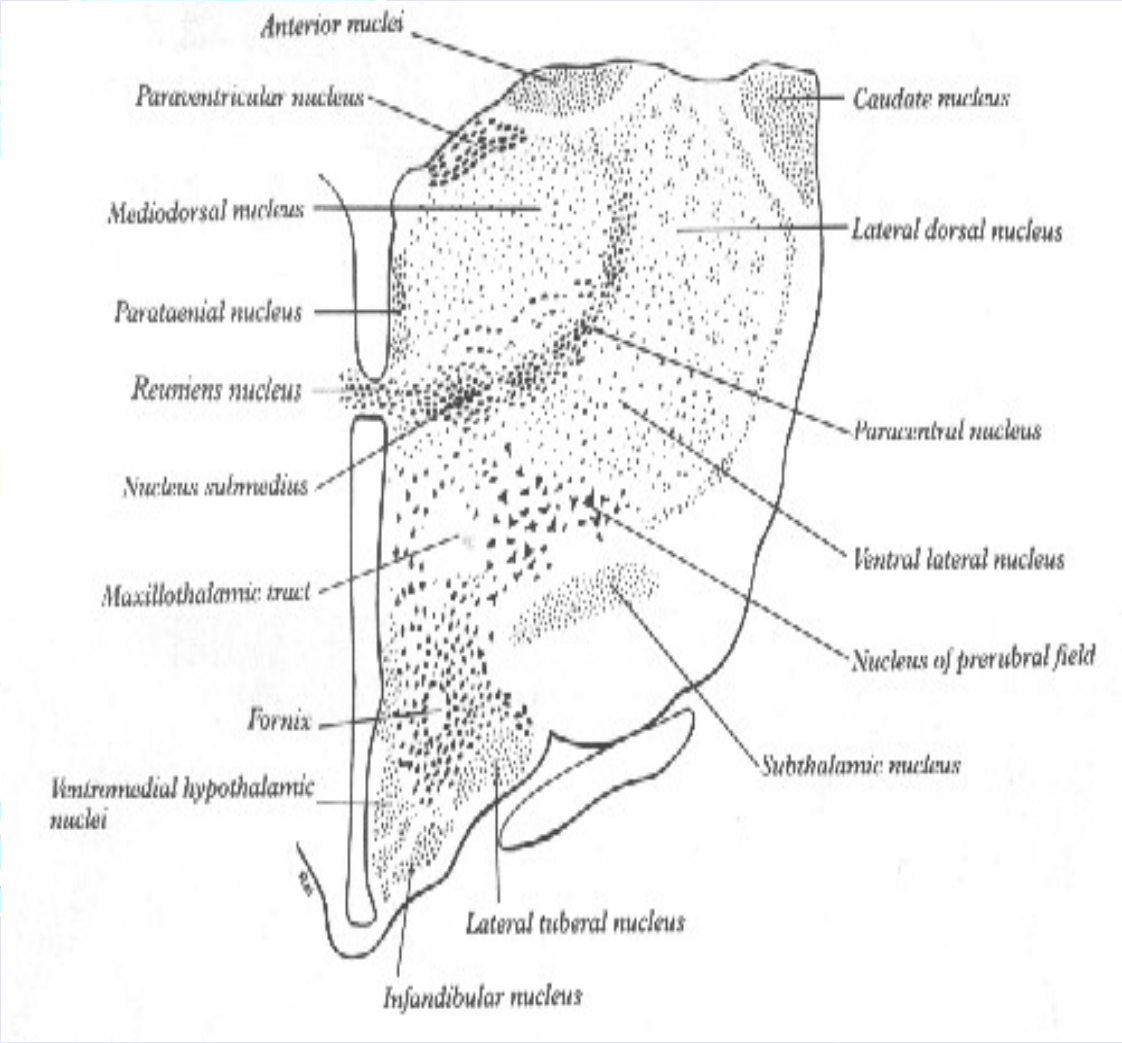
- AFFERENTS
  - **Superior colliculus**
- EFFERENTS
  - **Parietotemporal cortex**
  - **Visual area**
  - **Posterior temporal lobes**



(d) Transverse section of right side of thalamus showing locations of thalamic nuclei

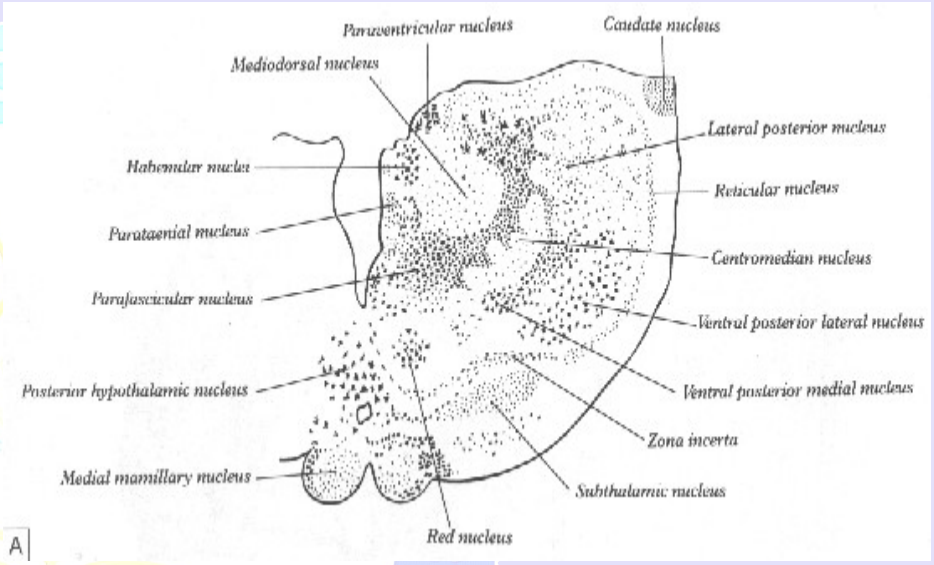
MIDLINE NUCLEI

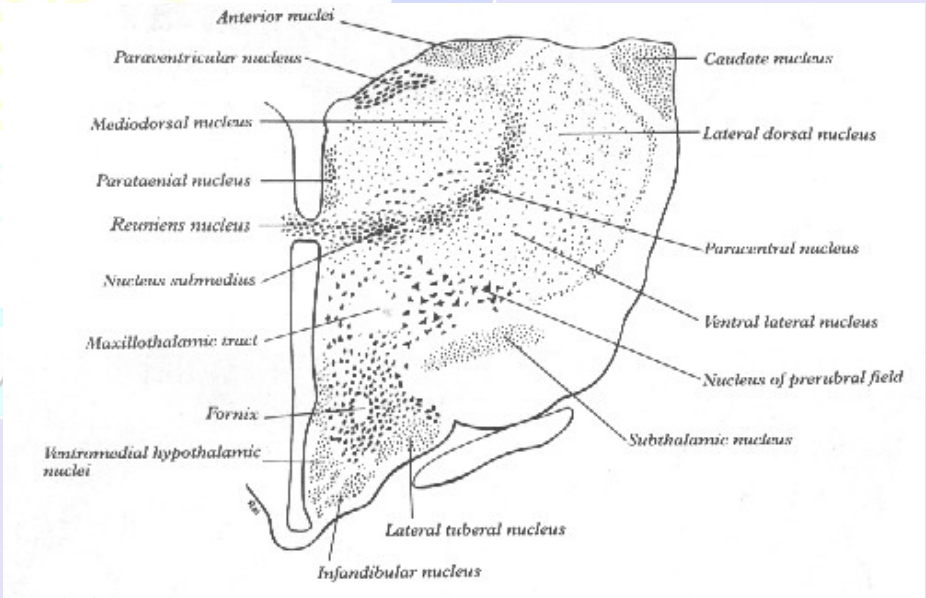
- **NUCLEI**
  - Paratenial- SC
  - Paraventricular- SC
  - Reuniens- SC
  - Rhomboidal- SC
  - Median central- SC
- **AFFERENTS**
- Hypothalamus periaqueductal gray, spinothalamic tract, reticular formation
- **EFFERENTS**
  - Amygdaloid nuclear complex
  - Ventral parts of striatum
  - Cingulate cortex



INTRALAMINAR NUCLEI

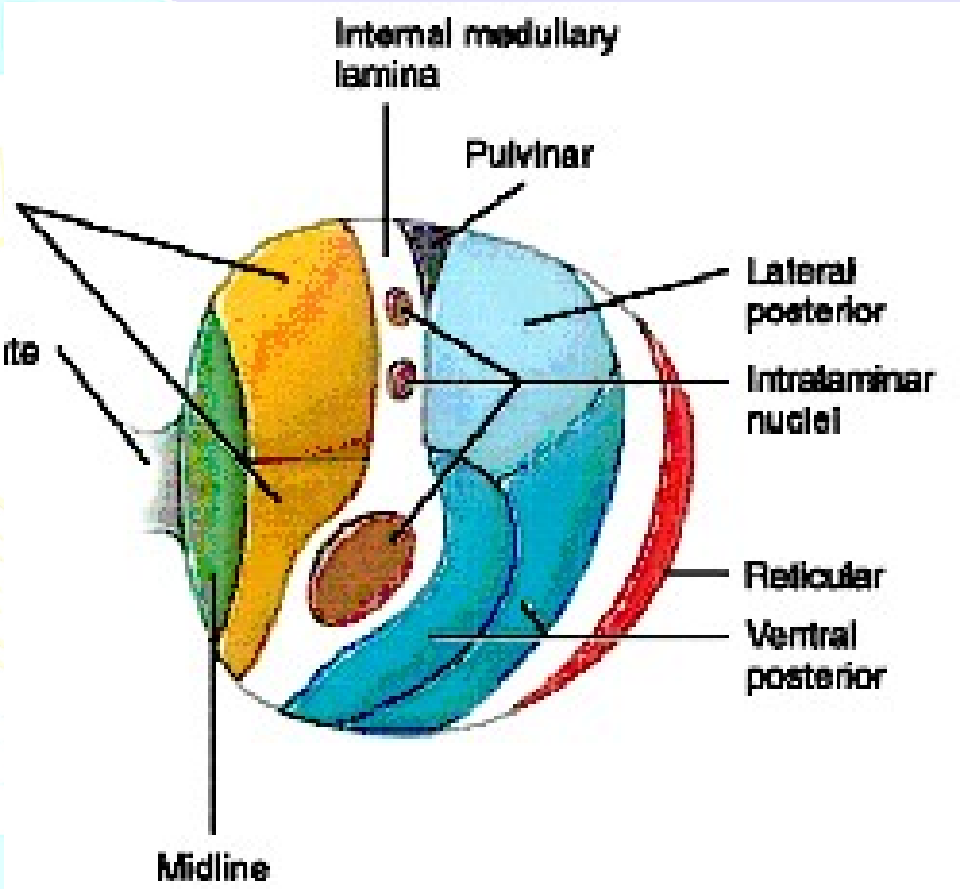
- **ROSTRAL DIVISION- SC, DC**
  - **Paracentral**
  - **Central lateral nucleus**
  - **Central median nucleus**
- **CAUDAL DIVISION- SC, DC**
  - **Centromedian nucleus-**
  - **Parafascicular nucleus**





INTRALAMINAR NUCLEI

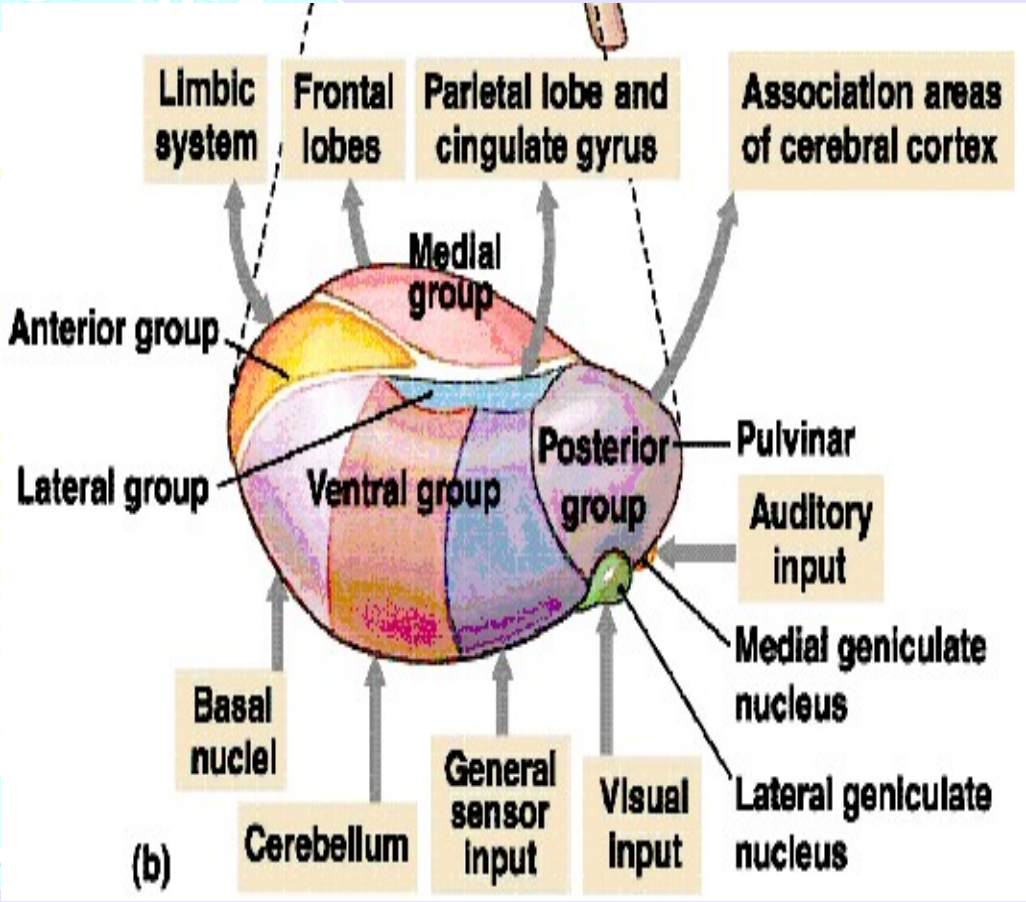
- AFFERENTS
  - Striatum
  - Substantia nigra
  - Brain stem reticular formation
  - Superior colliculus
  - Spinothalamic tract
  - Pretectal nuclei
- EFFERENTS
  - Widespread cortical areas
- FUNCTIONS
  - Mediates cortical activation from the brain stem reticular formation
  - Role in sensorimotor integration



(d) Transverse section of right side of thalamus showing locations of thalamic nuclei

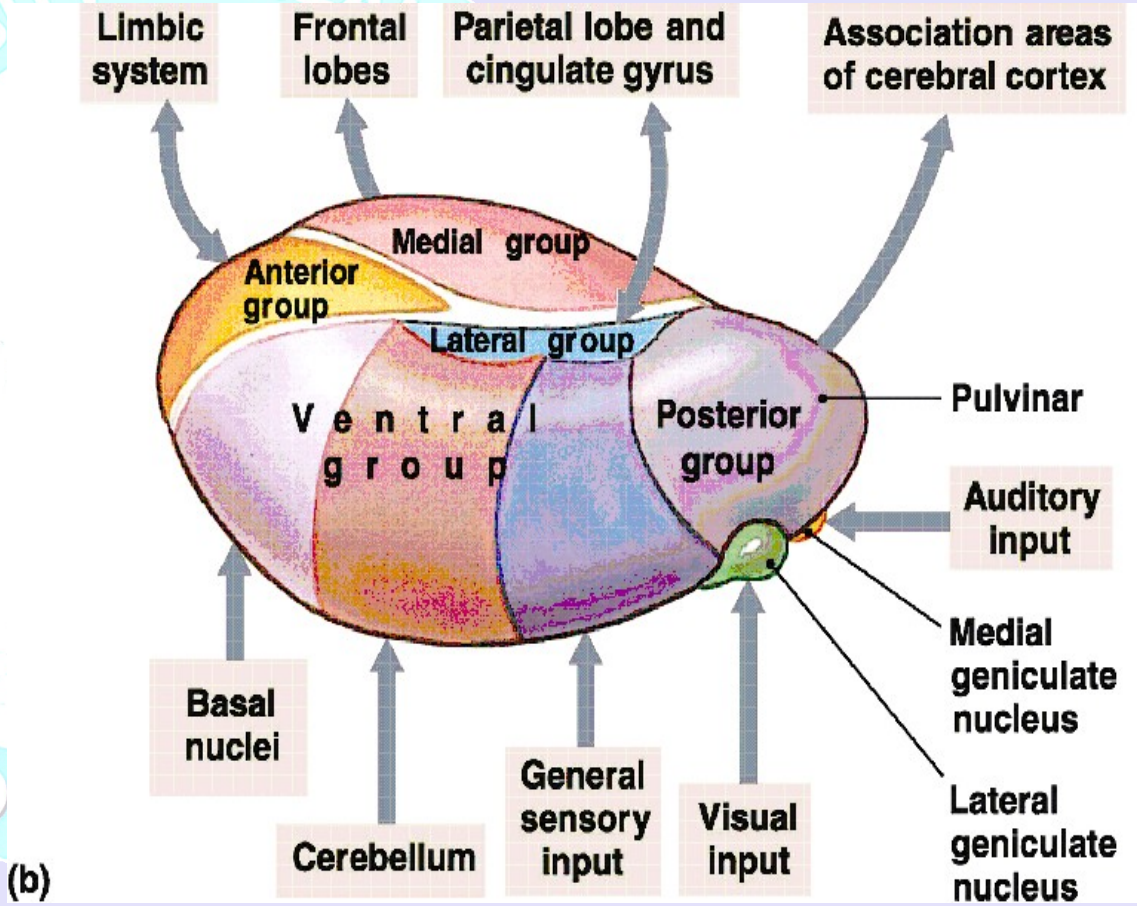
POSTERIOR NUCLEAR GROUP

- NUCLEI
  - **Supragenicular nucleus**
  - **Limitans nucleus**
  - **Posterior nucleus**
- AFFERENTS
  - **Spinothalamic medial leminiscal fibres**
  - **Primary somato-sensory cortex**
  - **Superior colliculus – Supragenicular**
- EFFERENTS
  - **Insular cortex**
  - **Auditory cortex**
- FUNCTIONS
  - **Integrate information of visual, auditory and somesthetic type**



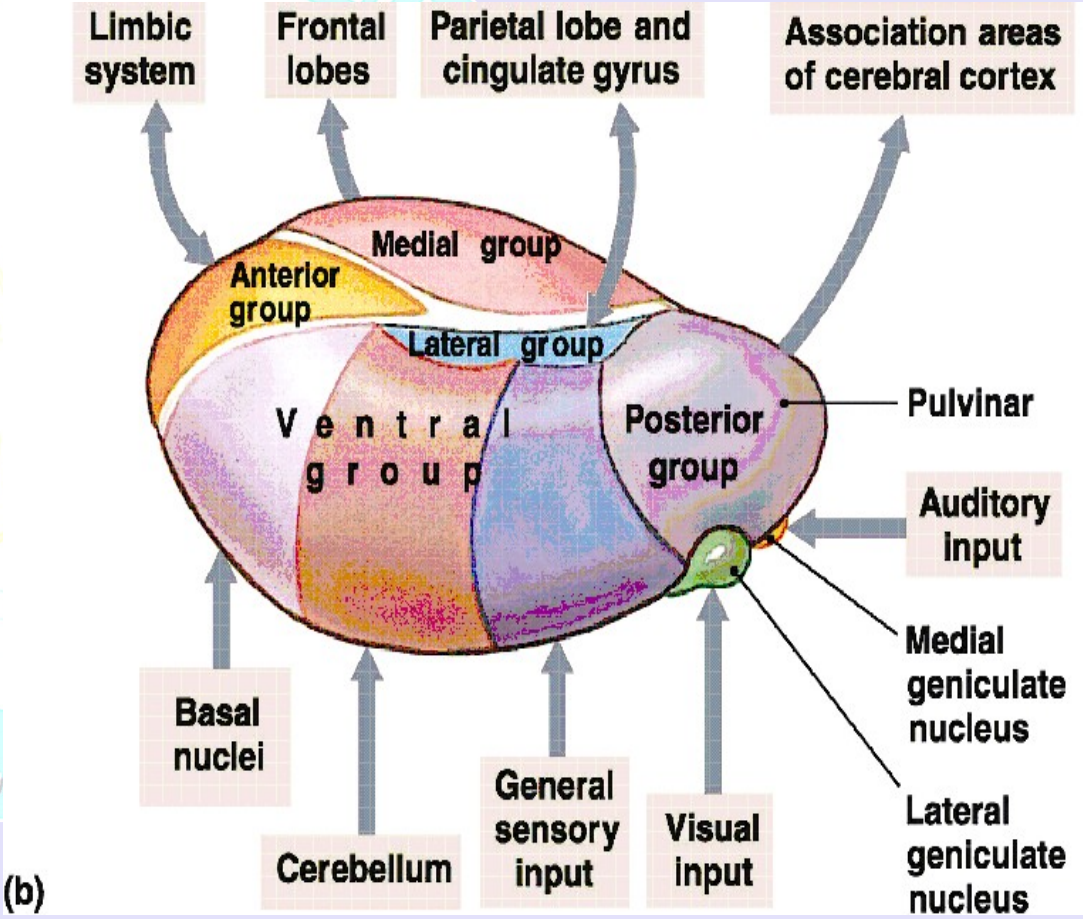
GENICULATE NUCLEAR GROUP

- **MEDIAL GENICULATE NUCLEUS**
  - **Parvicellular part**
  - **Magnocellular part**
  - **Dorsal part**
- **AFFERENTS**
  - **Auditory fibres from inferior colliculus via inferior brachium**
- **EFFERENTS**
  - **To primary auditory cortex via auditory radiations**



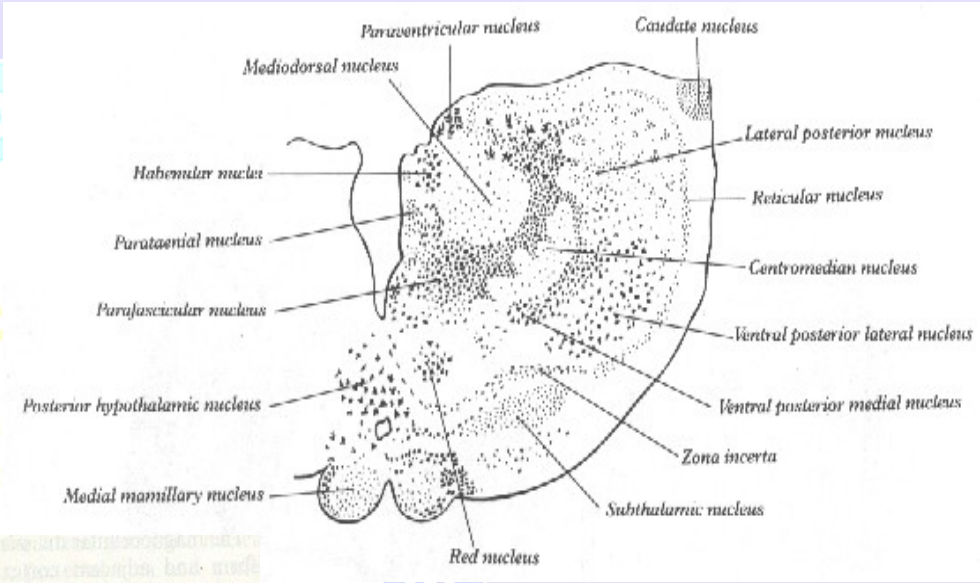
GENICULATE NUCLEAR GROUP

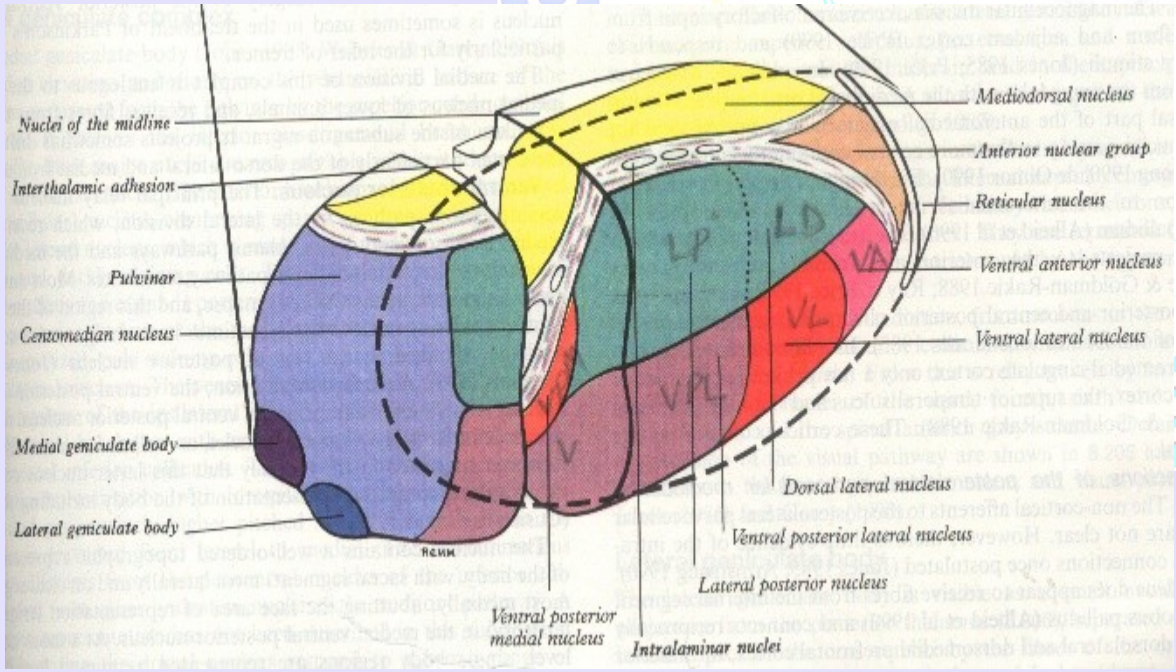
- **LATERAL GENICULATE NUCLEUS**
  - **Dorsal part**
  - **Ventral part**
- AFFERENTS
  - **Optic tract**
  - **Superior colliculus**
- EFFERENTS
  - **Primary visual cortex**
  - **Extra-striate visual areas**



VENTRAL THALAMUS

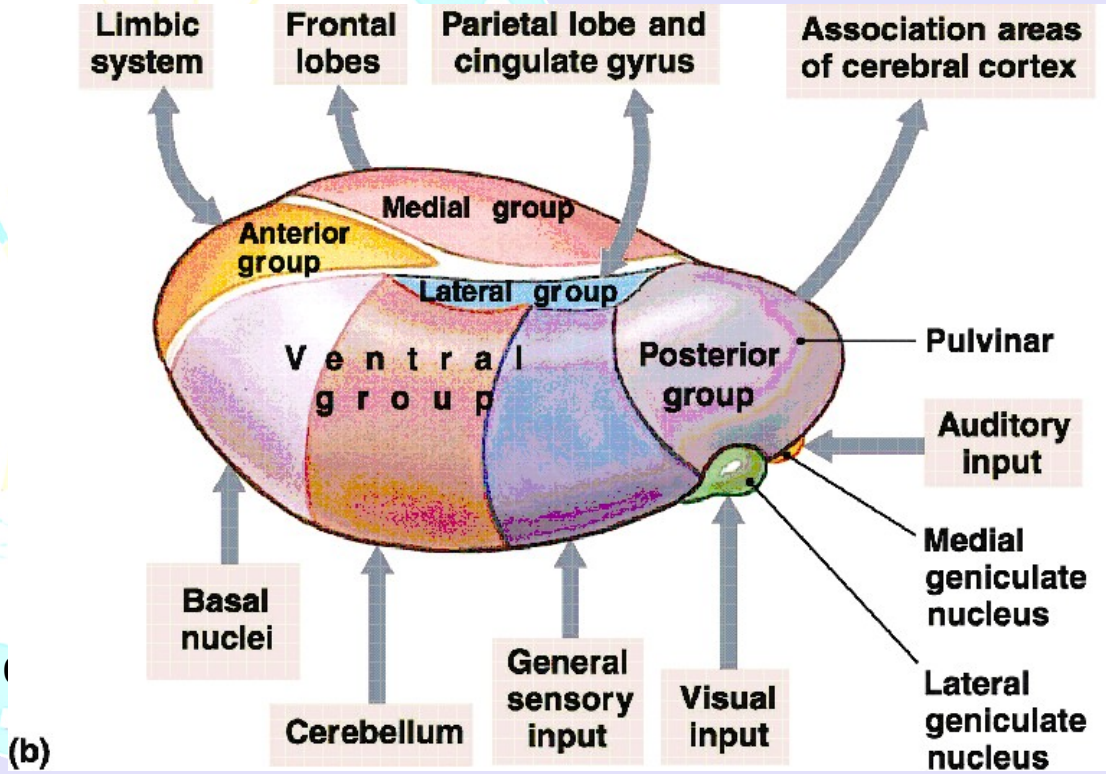
- **NUCLEI**
  - **Reticular nucleus**
  - **Pre-geniculate nucleus**
  - **Zona incerta**
  - **Fields of Forel**





VENTRAL THALAMUS CONNECTIONS

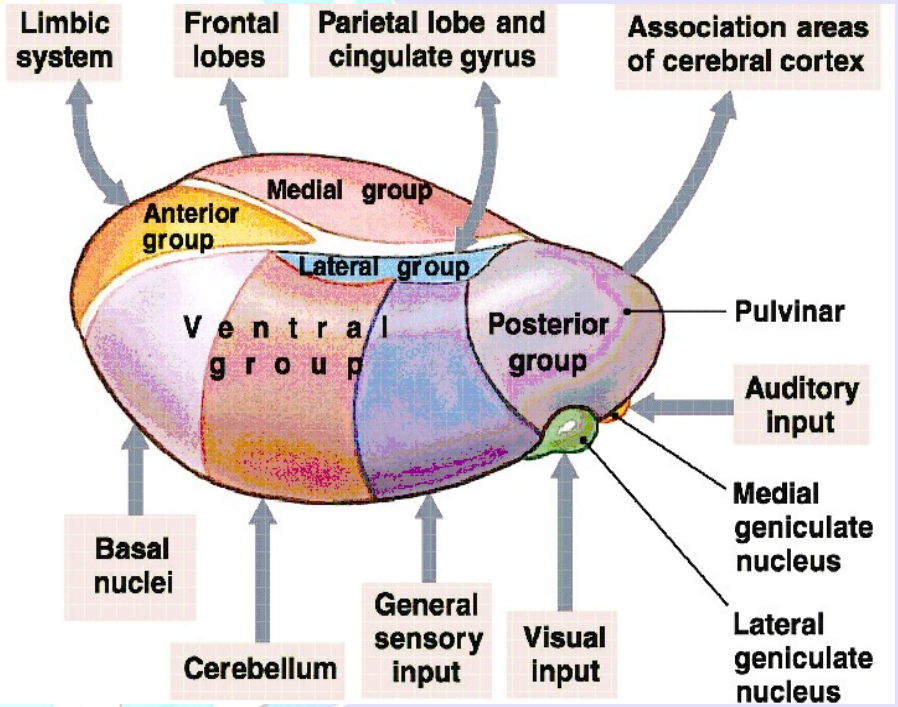
- **PREGENICULATE NUCLEUS**
  - AFFERENTS
    - **Retina**
    - **Visual cortex**
    - **Pretectum**
    - **Superior colliculus**
    - **Vestibular nucleus**
  - EFFERENTS
    - **Superior colliculus**
    - **Pretectum**
    - **Pontine nuclei**
    - **Hypothalamus**
  - FUNCTION
    - **Role in visuosensory an oculomotor processes**



VENTRAL THALAMUS CONNECTIONS

- **ZONA INCERTA**
  - AFFERENTS – ZI
    - **Sensorimotor cortex**
    - **Pregeniculate nucleus**
    - **Trigeminal nuclear complex**
    - **Spinal cord**
  - EFFERENTS
    - **Spinal cord**
    - **Pretectal region**

- **FIELDS OF FOREL**
  - **AFFERENTS**
    - **Globus Pallidus**
    - **Spinal cord**
    - **Brain stem reticular formation**
  - **EFFERENTS**
    - **Spinal cord**





THANK YOU